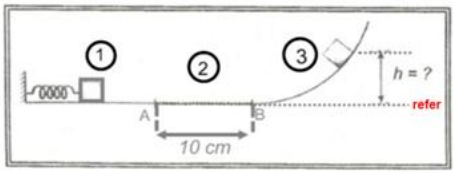
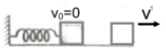
Componente Curricular: **Física – Dinâmica**
Professora: Rosana C. - Pádua **Série:** 2ª **Data:** Junho
Conteúdo: Sistemas Dissipativos – Exercícios - Resolução

1-



a) ① Sistema Conservativo



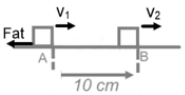
$E_{pe} = E_{cA}$

$\frac{k.x^2}{2} = E_{cA}$

$\frac{200.(0,1)^2}{2} = E_{cA}$

$E_{cA} = 1\ J$

b) ② O sistema não é conservativo



★ $\tau_{Fat} = -\ Fat.d$

$\tau_{Fat} = -\ 5.0,1$

$\tau_{Fat} = -\ 0,5\ J$

★ Fat = μ.N

Fat = 0,5.10

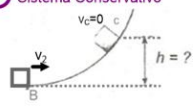
Fat = 5 N

$\tau_{Fat} = E_{CB} - E_{CA}$

$-0,5 = E_{CB} - 1$

$E_{CB} = 0,5\ J$

c) ③ Sistema Conservativo



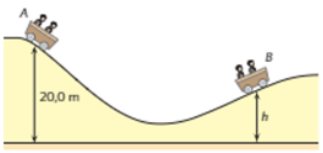
$E_{CB} = E_{pC}$

$E_{cB} = m.g.h$

$0,5 = 1.10.h$

$h = 0,05\ m \Rightarrow h = 5\ cm$

2-



$E_{dis} = E_{MA} - E_{MB}$

$E_{dis} = m.g.h_A - m.g.h_B$

$4,55.10^4 = 6,5.10^3.20 - 6,5.10^3.h$

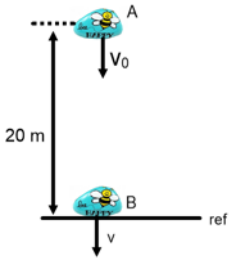
$4,55.10^4 = 13.10^4 - 0,65.10^4.h$

$(4,55 - 13).10^4 = 0,65.10^4.h$

$h = \frac{8,45}{0,65} \Rightarrow h = 13\ m$

3-

$v_0 = 10\ m/s$
$v = 20\ m/s$



$E_{térmica} = E_{dissipada}$

A: $E_{MA} = E_{CA} + E_{PA}$

$E_{MA} = \frac{m.v_0^2}{2} + mgh \longrightarrow E_{MA} = \frac{0,6.10^2}{2} + 0,6.10.20$

$E_{MA} = 150\ J$

** $E_{MB} = E_{CB} + E_{pB}$

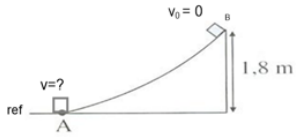
$E_{MB} = \frac{m.v^2}{2} \longrightarrow E_{MB} = \frac{0,6.20^2}{2} \longrightarrow E_{MB} = 120\ J$

$E_{dis} = E_{MA} - E_{MB}$

$E_{dis} = 150 - 120 \longrightarrow E_{dis} = 30\ J$

*$\tau_{Fdis} = -\ 30\ J$

4-



a) $E_{MB} = E_{MA}$

$$m.g.h = \frac{m.v_A^2}{2}$$

$$v_A^2 = 2.g.h$$

$$v_A^2 = 2.10.1,8 \rightarrow \quad v_A^2 = 36$$

v_A = 6 m/s

b) $E_{MB} = m.g.h$

$E_{MB} = 8.10.1,8 \rightarrow E_{MB} = 144$ J

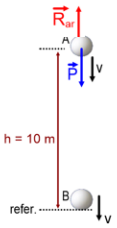
$$E_{MA} = \frac{m.v_A^2}{2}$$

$$E_{MA} = \frac{8.4^2}{2} \rightarrow E_{MA} = 64 \text{ J}$$

$E_{dis} = E_{MB} - E_{MA}$

$E_{dis} = 144 - 64 \rightarrow E_{dis} = 80$ J

5-



A: $E_{MA} = E_{CA} + E_{PA} \Rightarrow E_{MA} = \frac{m.v^2}{2} + m.g.h$

B: $E_{MB} = E_{CB} \Rightarrow E_{MB} = \frac{m.v^2}{2}$

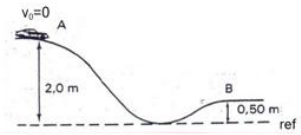
$E_{dis} = E_{MA} - E_{MB}$

$E_{dis} = m.g.h$

$E_{dis} = 2,5.10^{-3}.10.10 \Rightarrow E_{dis} = 0,25$ J

6-

*$E_{dis} = 40\%.E_{MA}$



$E_{MB} = 0,6.E_{MA}$

$E_{CB} + E_{PB} = 0,6.E_{pA}$

$$\frac{m.v_B^2}{2} + m.g.h_B = 0,6.m.g.h_A$$

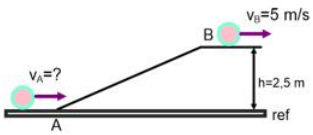
$$v_B^2 + 2.g.h_B = 1,2.g.h_A$$

$$v_B^2 + 2.10.0,5 = 1,2.10.2$$

$$v_B^2 = 14 \rightarrow v_B = \sqrt{14}\, m/s$$

7-

*$E_{dis} = 25\%.E_{MA}$



$E_{MB} = 75\%.E_{MA}$

$E_{MB} = 0,75.E_{MA}$

$E_{CB} + E_{PB} = 0,75.E_{cA}$

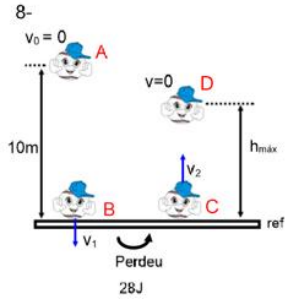
$$\frac{m.v_B^2}{2} + m.g.h_B = 0,75.\frac{m.v_A^2}{2}$$

$$v_B^2 + 2.g.h_B = 0,75.v_A^2$$

$$5^5 + 2.10.2,5 = 0,75.v_A^2$$

$$v_A^2 = 100 \rightarrow v_A = 10\, m/s$$

8-



★ A-B: Sistema Conservativo

$E_{MA} = E_{MB}$

$E_{pA} = E_{MB}$

$E_{MB} = m.g.h_A$

★ C-D: Sistema Conservativo

$E_{MC} = E_{MD}$

$E_{MC} = E_{pD}$

$E_{MC} = m.g.h_{máx}$

★ B-C: Sistema Dissipativo

$E_{dis} = E_{MB} - E_{MC}$

$E_{dis} = m.g.h_A - m.g.h_{máx}$

$28 = 1.10.10 - 1.10.h_{máx}$ => $28 = 100 - 10.h_{máx}$

$10.h_{máx} = 72$

$h_{máx} = 7,2$ m